

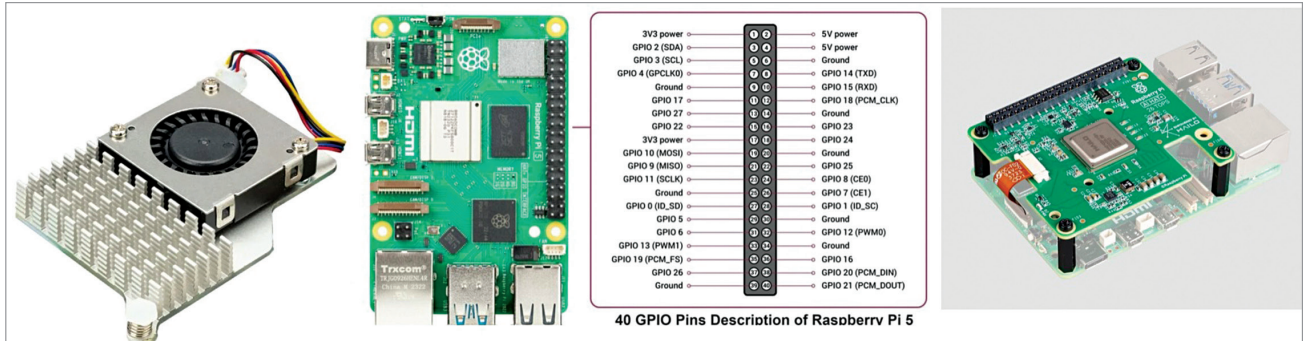


Fig. 2: AI accelerator mounted on Raspberry Pi

```
$ ssh bera@192.168.x.x
```

When prompted, enter the password. Once authenticated, access to the Raspberry Pi 5 terminal will be granted.

Note. SSH access is optional; a direct connection using a display, keyboard and mouse is also supported.

Enabling PCIe Gen 3.0 speeds. By default, the Raspberry Pi 5 operates at Gen 2.0 PCIe speeds. To enable Gen 3.0:

```
$ sudo raspi-config
```

- Navigate to Advanced Options →PCIe Speed
- Choose 'Yes' to enable Gen 3 mode
- Select 'Finish,' and when prompted, reboot the system.

Install Hailo software stack. After rebooting, install the Hailo drivers and tools:

```
$ sudo apt install hailo-all
```

This command installs the Hailo kernel device driver, firmware, HailoRT, Hailo applications, libraries, and `rplicam-apps`. Then reboot the system:

```
$ sudo reboot
```

Reboot to apply all changes. Verify Hailo Installation. After reboot, run:

```
$ hailortcli fw-control identify
```

This ensures all changes take effect.

```
Executing on device: 0000:01:00.0
Identifying board
Control Protocol Version: 2
```

```
Firmware Version: 4.17.0
```

```
(release,app,extended context switch
buffer) Logger Version: 0
```

```
Board Name: Hailo-8
```

```
Device Architecture: HAILO8L
```

```
Serial Number: HLDDLBB234500054
```

```
Part Number: HM21LB1C2LAE
```

```
Product Name: HAILO-8L AI ACC M.2 B+M KEY
MODULE EXT TMP
```

The last five lines ensure that the Hailo-8L board has been successfully installed. In some instances, the serial or part number may not be displayed, but the board will continue to function correctly. Test rplicam-hello:

```
$> rplicam-hello -t 10s # This will run
rplicam for 10 seconds and then stop
$> rplicam-hello --help # This displays
the help file
```

Once all components are installed, install the `rplicam-apps` package to enable Hailo AI accelerator features:

```
$> sudo apt update && sudo apt install
rplicam-apps
```

High-speed object detection examples can now be executed:

```
$> rplicam-hello -t 0 --post-process-file /
usr/share/rpi-camera-assets/hailo_yolov6_
inference.json
$> rplicam-hello -t 0 --post-process-file /
usr/share/rpi-camera-assets/hailo_yolov8_
inference.json
$> rplicam-hello -t 0 --post-process-file /
usr/share/rpi-camera-assets/hailo_yolov5_
personface.json # This command detects
faces - useful for face counting
$> rplicam-hello -t 0 --post-process-file /
```

```
usr/share/rpi-camera-assets/hailo_yolov8_
pose.json # This command detects human
poses - useful for fall detection
```

These commands demonstrate how to use `rplicam-hello` with different YOLO AI models.

Installation instructions. Hailo RPi5 Examples - Installation Guide:

Step 1: Clone the repository

```
$> git clone https://github.com/hailo-ai/
hailo-rpi5-examples.git
```

Step 2: Navigate into the directory

```
$> cd hailo-rpi5-examples
```

This directory contains two key files: `setup\_env.sh` and `requirements.txt`.

As the Hailo-8L requires specific Python package versions, it is advisable to use a virtual environment to avoid conflicts. This allows for the installation of the necessary package versions locally without interfering with the system-wide Python environment. The next step is to activate the special environment as recommended by the Hailo-8L package.

Step 3: Activate the virtual environment

```
$> source setup_env.sh
```

Once activated, the terminal prompt will change to indicate entry inside the virtual environment: (venv_hailo_rpi5_examples) bera@raspberrypi: ~/hailo-rpi5-examples \$

Step 4: Install required Python packages

```
(venv_hailo_rpi5_examples)
bera@raspberrypi: ~/hailo-
rpi5-examples $ pip install -r
```